

# Windows Endpoint Persistence Investigation

**Scenario:** A suspicious executable named challenge.exe has been executed on a Windows workstation. Your task is to identify listening network activity and persistence mechanisms created by the binary using endpoint analysis commands.

**Lab safety note:** This is a controlled classroom challenge. Run it only inside the approved lab VM as discussed during the lecture. Keep the challenge window open until the investigation is complete, then use the revert command.

**Student task:** Complete each question. For each answer, submit the command used, the value found, and a brief justification explaining how the command supports your answer.

**Start the Challenge:** Open Command Prompt as Administrator. Navigate to the folder containing challenge.exe, then execute it. **Leave the window open until the investigation is complete.**

challenge.exe

Open a second Administrator PowerShell window for the investigation. Do not run the investigation in the same window where challenge.exe is running.

**Start the challenge** - Confirm that challenge.exe has been executed and is still running.

<Provide PowerShell command(s) here>

**Identify the malware process ID:** Find the PID of challenge.exe.

<Provide PowerShell command(s) here>

**Identify the listening port** - Find the local TCP port opened by challenge.exe.

<Provide PowerShell command(s) here>

**Identify the executable path:** Determine the full path from which challenge.exe is running.

<Provide PowerShell command(s) here>

**List DLL modules beginning with m:** List loaded modules for challenge.exe and identify the two DLLs whose names begin with m.

<Provide PowerShell command(s) here>

**Identify the parent process:** Determine which process launched challenge.exe.

<Provide PowerShell command(s) here>

**Find the attacker-created SMB share:** List local SMB shares and identify the non-default share created by the binary.

<Provide PowerShell command(s) here>

**Identify the SMB share path:** Determine the local folder path exposed by the attacker-created share. Replace <ShareName> with the suspicious share name.

<Provide PowerShell command(s) here>

**Identify the Registry Run key path:** Find the Registry Run key used by the binary for user logon persistence. Look for a suspicious value pointing to a user-writable path such as Downloads, Documents, AppData, or Temp.

<Provide PowerShell command(s) here>

**Identify the malicious Run entry name:** Find the name of the suspicious Registry value in the Run key.

<Provide PowerShell command(s) here>

**Identify the Run entry image path:** Find the executable path configured in the suspicious Run key value. Replace <EntryName> with the suspicious value name.

<Provide PowerShell command(s) here>

**Identify the backdoor service name:** Find the suspicious Windows service installed by the binary. Some ProgramData services may be legitimate. Focus on unusual services pointing to user folders.

<Provide PowerShell command(s) here>

**Identify the service start type:** Determine whether the suspicious service is set to start automatically or manually. Use sc.exe in PowerShell, not sc alone.

<Provide PowerShell command(s) here>

**Identify the service binary path:** Find the executable path configured as the service binary.

<Provide PowerShell command(s) here>

**Identify the attacker-created scheduled task:** Find the scheduled task created by the binary. Ignore known legitimate Windows Defender tasks under C:\ProgramData\Microsoft\Windows Defender.

<Provide PowerShell command(s) here>

**Identify the scheduled task executable:** Determine the executable launched by the suspicious scheduled task.

<Provide PowerShell command(s) here>

**Identify the scheduled task run time:** Determine when the scheduled task is configured to run. Look for StartBoundary or Start Time.

<Provide PowerShell command(s) here>

**Clean up and revert the system:** Terminate challenge.exe and restore the system to its original state. Enter done after cleanup is complete.

<Provide PowerShell command(s) here>

### Submission Requirements

- Answers to all questions.
- The command used for each technical answer.
- A brief explanation of the evidence.
- Screenshots are optional unless requested by the instructor.

**Complete the following table:**

The answers below are based on the specific execution output provided for this lab run.

| Question | Finding                    | Expected answer |
|----------|----------------------------|-----------------|
| 1        | Challenge started          | YES             |
| 2        | PID of challenge.exe       |                 |
| 3        | Listening port             |                 |
| 4        | Full path of challenge.exe |                 |
| 5        | DLLs beginning with m      |                 |
| 6        | Parent process             |                 |
| 7        | Attacker-created SMB share |                 |
| 8        | Share path                 |                 |
| 9        | Registry Run key path      |                 |
| 10       | Malicious Run entry name   |                 |
| 11       | Run key image path         |                 |
| 12       | Backdoor service name      |                 |
| 13       | Service start type         |                 |
| 14       | Service binary path        |                 |
| 15       | Scheduled task name        |                 |
| 16       | Scheduled task executable  |                 |
| 17       | Scheduled task run time    |                 |